Valence Alignment in Human-AI Story Co-Creation

Analysis of Field Data and AI-AI Baseline Comparison

PENPAL Project

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0.1 Summary

- Emotional alignment is asymmetric: the model reliably aligns to the user (User → AI), while humans do not mirror the preceding AI tone.
- AI–AI interactions show higher, bidirectional alignment driven by symmetric context access.
- Field data shows greater variance in valence and weaker overall coordination compared to the simulated dataset.
- Participant-level modelling reveals a consistent "rubber-band" effect, where early divergence returns toward baseline in later stages.
- Overall, affective dynamics in the field are primarily model-driven rather than mutually convergent.

1 1. Data Loading and Preprocessing

1.1 1.1 Load Data

```
## Field data dimensions: 820 14

## Simulated data dimensions: 30 12

##
## Field data columns:

##  [1] "timestamp"            "user"                 "ai"
##  [4] "conversation_id"      "respondent_id"        "interaction_count"
##  [7] "llm_type"             "turn"                 "starter"
## [10] "user_embedding"       "ai_embedding"         "pct_turn"
## [13] "user_sentiment_score" "ai_sentiment_score"

##
## Simulated data columns:

##  [1] "turn"                 "user"                 "ai"
##  [4] "client_id"            "workshop_id"          "conversation_id"
##  [7] "timestamp"            "user_embedding"       "ai_embedding"
## [10] "pct_turn"             "user_sentiment_score" "ai_sentiment_score"
```

1.2 1.2 Data Preprocessing

```
##
## === Field Data Summary ===

## # A tibble: 1 x 5
##   n_conversations n_turns mean_turns_per_story mean_user_valence mean_ai_valence
##             <int>   <int>                <dbl>             <dbl>           <dbl>
## 1              81     729                    9            0.0426           0.171

##
## === Simulated Data Summary ===

## # A tibble: 1 x 5
##   n_conversations n_turns mean_turns_per_story mean_user_valence mean_ai_valence
##             <int>   <int>                <dbl>             <dbl>           <dbl>
## 1               6      30                    5            -0.407          -0.296
```

1.3 1.3 Compute Lagged Variables and Cosine

```
## === Lag Variable Verification ===

## Field data - Turn 1 with non-NA lag (should be 0): 0

## Simulated data - Turn 1 with non-NA lag (should be 0): 0
```

2 2. Descriptive Statistics

2.1 2.1 Token Count Statistics

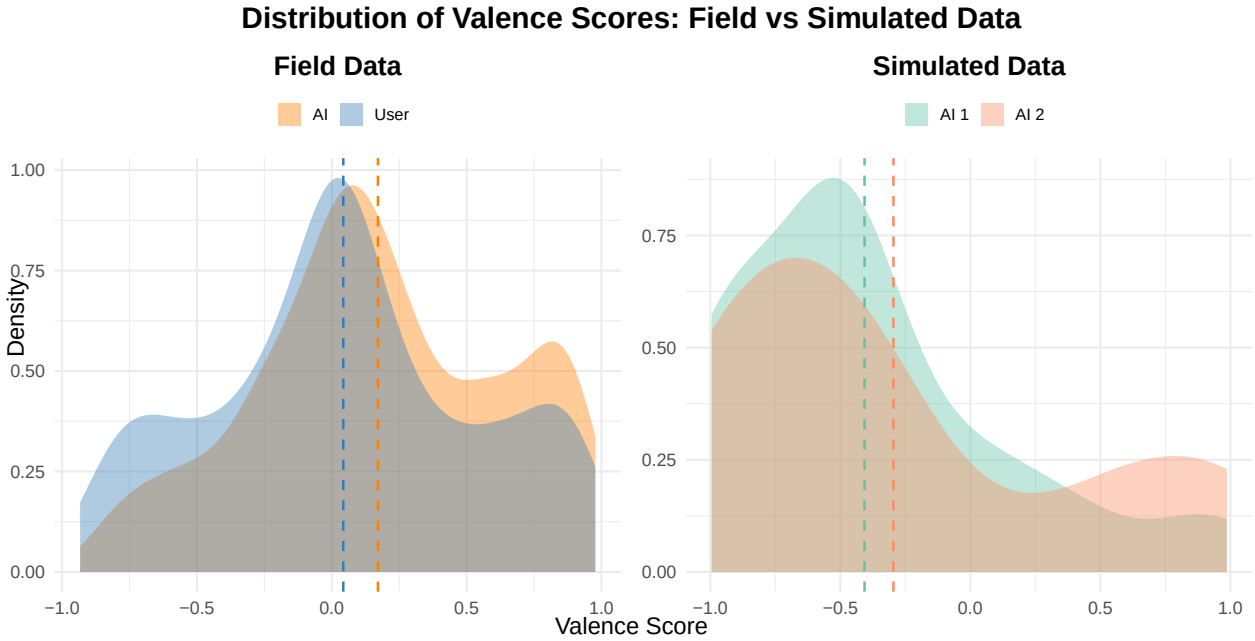
```
## === Field Data Token Statistics ===

## # A tibble: 1 x 5
##   mean_user_tokens sd_user_tokens mean_ai_tokens sd_ai_tokens mean_token_ratio
##              <dbl>          <dbl>          <dbl>        <dbl>            <dbl>
## 1             25.2           5.17           28.7         7.65            0.924

##
## === Simulated Data Token Statistics ===

## # A tibble: 1 x 5
##   mean_user_tokens sd_user_tokens mean_ai_tokens sd_ai_tokens mean_token_ratio
##              <dbl>          <dbl>          <dbl>        <dbl>            <dbl>
## 1             26.0           2.17           41.1         3.73            0.636
```

2.2 2.2 Valence Distribution Statistics

```
## # A tibble: 4 x 5
##   data_source agent mean_valence sd_valence median_valence
##   <chr>       <chr>        <dbl>      <dbl>          <dbl>
## 1 Field       ai          0.171       0.460         0.137
## 2 Field       user        0.0426      0.492         0.0239
## 3 Simulated   ai         -0.296       0.681        -0.487
## 4 Simulated   user       -0.407       0.529        -0.504
```

3 3. Valence Distribution Analysis

3.1 3.1 Combined Valence Distribution Plot



4 4. Valence Alignment Analysis

4.1 4.1 Per-Story Correlation (Same Turn)

```
## === Same-Turn Alignment Summary ===
```

```
## # A tibble: 2 x 5
##   data_source mean_rho sd_rho median_rho n_stories
##   <chr>          <dbl>  <dbl>      <dbl>     <int>
## 1 Field          0.259  0.351      0.309        81
## 2 Simulated     -0.246  0.440     -0.314         6
```

4.2 4.2 Per-Story Correlation (Previous Turn)

```
## === Previous-Turn Alignment Summary ===
```

```
## # A tibble: 2 x 5
##   data_source mean_rho sd_rho median_rho n_stories
##   <chr>          <dbl>  <dbl>      <dbl>     <int>
## 1 Field          0.109  0.355     0.0669        81
## 2 Simulated      0.533  0.224     0.546          6
```

4.3 4.3.1 All turns story correlation (same turn)

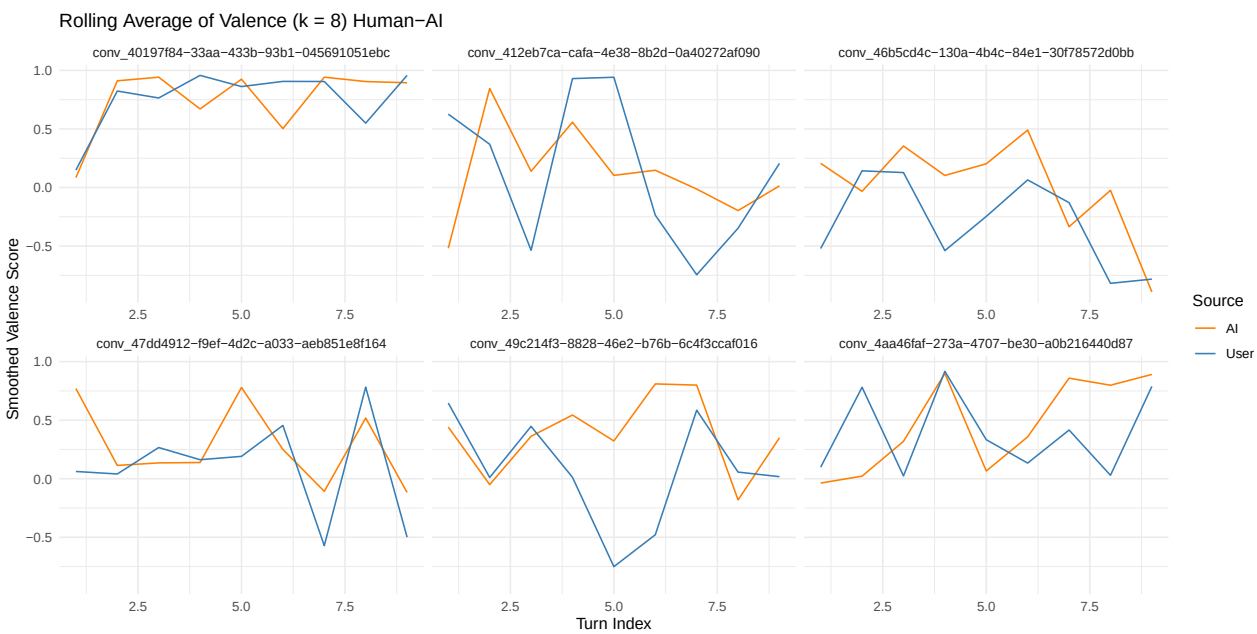
```
## === Same-Turn All Turns Alignment Summary ===
```

```
## # A tibble: 2 x 5
##   data_source mean_rho sd_rho median_rho n_stories
##   <chr>          <dbl>  <dbl>      <dbl>     <int>
## 1 Field          0.395     NA      0.395         1
## 2 Simulated     -0.110     NA     -0.110         1
```

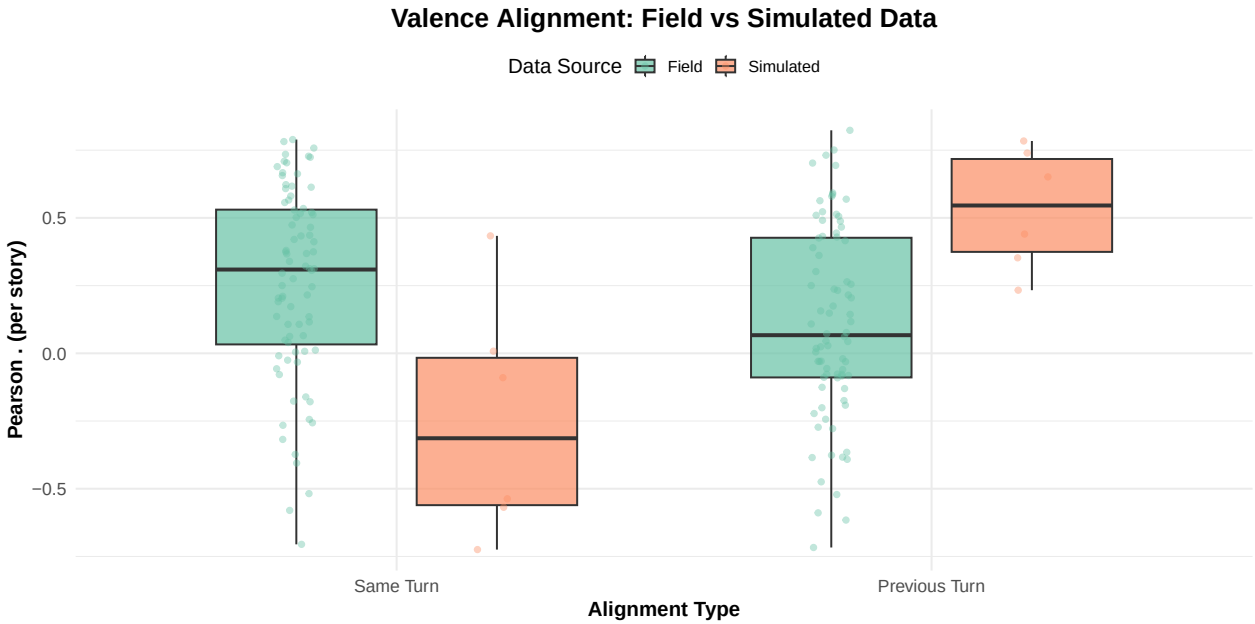
4.4 4.3.2 All turns story correlation (Previous Turn)

```
## === Previous-Turn Aggregate Correlation ===

## Field: Turn 1 observations (excluded): 0

## Field: Valid lag observations: 648

## Simulated: Turn 1 observations (excluded): 6

## Simulated: Valid lag observations: 24

## === Previous-Turn All Turns Alignment Summary ===

## # A tibble: 2 x 6
##   data_source mean_rho sd_rho median_rho n_total n_valid
##   <chr>          <dbl>  <dbl>      <dbl>   <int>   <int>
## 1 Field          0.311     NA      0.311     729     648
## 2 Simulated      0.425     NA      0.425      30      24
```

4.5 4.3.3 Probe Sentiment Narrative Development



4.6 4.4.1 Alignment Comparison Plot



4.7 4.4.2 Comparison: Per-Story vs Aggregate Correlations

```
##
## === Per-Story vs Aggregate Correlation Comparison ===
## # A tibble: 4 x 4
```

```
##   data_source alignment_type `Per-Story Mean` `Aggregate All Turns`
##   <chr>       <fct>                     <dbl>                 <dbl>
## 1 Field       Same Turn                 0.259                 0.395
## 2 Field       Previous Turn             0.109                 0.311
## 3 Simulated   Same Turn               -0.246                -0.110
## 4 Simulated   Previous Turn             0.533                 0.425
```

4.8 4.5 Mixed Effects Model: Random Intercepts per Story

```
## === Mixed Effects Model: Same-Turn Alignment (Field Data) ===

##               Estimate Std. Error        df  t value     Pr(>|t|)
## (Intercept)  0.1601309 0.02852669  76.98068 5.613371 3.004702e-07
## user_valence 0.2581102 0.03016732 710.75841 8.555952 7.167629e-17

##
## === Mixed Effects Model: Previous-Turn Alignment (Field Data) ===

##                     Estimate Std. Error        df   t value     Pr(>|t|)
## (Intercept)     -0.01547305 0.02477547  82.23112 -0.624531 5.340079e-01
## ai_valence_lag1  0.27743750 0.04273432 512.25953  6.492148 2.002318e-10
```

4.9 4.6 Statistical Tests: Alignment Differences

```
## === T-test: Same-Turn Alignment ===

##
##  Welch Two Sample t-test
##
## data:  rho by data_source
## t = 2.7469, df = 5.4867, p-value = 0.03667
## alternative hypothesis: true difference in means between group Field and group Simulated is not equal
## 95 percent confidence interval:
##  0.04477244 0.96558067
## sample estimates:
##     mean in group Field mean in group Simulated
##               0.2587301              -0.2464464

##
## === T-test: Previous-Turn Alignment ===

##
##  Welch Two Sample t-test
##
## data:  rho by data_source
## t = -4.2627, df = 7.0529, p-value = 0.003671
## alternative hypothesis: true difference in means between group Field and group Simulated is not equal
## 95 percent confidence interval:
##  -0.6602888 -0.1895659
## sample estimates:
##     mean in group Field mean in group Simulated
##               0.1085075               0.5334349

##
## === One-Sample Tests Against Zero ===

## # A tibble: 4 x 6
##   data_source alignment_type t_stat       p_value mean_rho     n
```

```
##   <chr>       <fct>           <dbl>         <dbl>    <dbl> <int>
## 1 Field       Same Turn        6.60 0.00000000432    0.259    81
## 2 Field       Previous Turn    2.73 0.00781          0.109    81
## 3 Simulated   Same Turn       -1.37 0.229           -0.246     6
## 4 Simulated   Previous Turn    5.84 0.00209          0.533     6
```

5 5. Summary and Conclusions

5.1 5.1 Key Findings Summary

```
## # A tibble: 5 x 3
##   Metric                      Field_Data Simulated_Data
##   <chr>                            <dbl>          <dbl>
## 1 N Stories                       81              6
## 2 Mean User Valence                0.043         -0.407
## 3 Mean AI Valence                  0.171         -0.296
## 4 Same-Turn Alignment ( )         0.259         -0.246
## 5 Previous-Turn Alignment ( )      0.109          0.533
```

6 6. Supplementary: Interaction Effects

6.1 6.1 Starter Interaction Effects

6.1.1 6.1.1 Same-Turn Alignment by Starter

```
##                      Estimate  Std. Error        df  t value   Pr(>|t|)
## (Intercept)       0.064636123 0.037711094  270.9481 1.713982 0.08767574
## interaction_count 0.007036409 0.004301995 1376.0000 1.635615 0.10214884
```

```
## === Same-Turn Alignment: Starter Interaction ===
```

```
##                              Estimate Std. Error        df     t value
## (Intercept)               0.15746815 0.04047342  75.50867  3.89065643
## user_valence              0.27254491 0.04461448 709.98874  6.10888886
## starteruser               0.00567157 0.05762066  75.63778  0.09842944
## user_valence:starteruser -0.02804765 0.06060062 708.73272 -0.46282780
##                              Pr(>|t|)
## (Intercept)              2.136849e-04
## user_valence             1.652752e-09
## starteruser              9.218517e-01
## user_valence:starteruser 6.436299e-01
```

6.1.2 6.1.2 Previous-Turn Alignment by Starter

```
## === Previous-Turn Alignment: Starter Interaction ===
```

```
##                                Estimate Std. Error        df    t value
## (Intercept)                 -0.01949332 0.03434770  82.65992 -0.5675290
## ai_valence_lag1              0.16726461 0.06229103 539.00441  2.6852116
## starteruser                  0.01028729 0.04868656  81.32617  0.2112964
## ai_valence_lag1:starteruser  0.21102580 0.08511384 503.72190  2.4793359
##                                Pr(>|t|)
## (Intercept)                 0.571893032
## ai_valence_lag1             0.007471671
## starteruser                 0.833184605
## ai_valence_lag1:starteruser 0.013488802
```

6.1.3 6.1.3 Per-Story Correlations by Starter

```
## === Same-Turn Correlations by Starter ===
```

```
## # A tibble: 2 x 4
##   starter mean_rho sd_rho n_stories
##   <chr>      <dbl>  <dbl>     <int>
## 1 ai         0.285  0.372        41
## 2 user       0.232  0.330        40
```

```
##
## T-test (starter effect on same-turn rho):
```

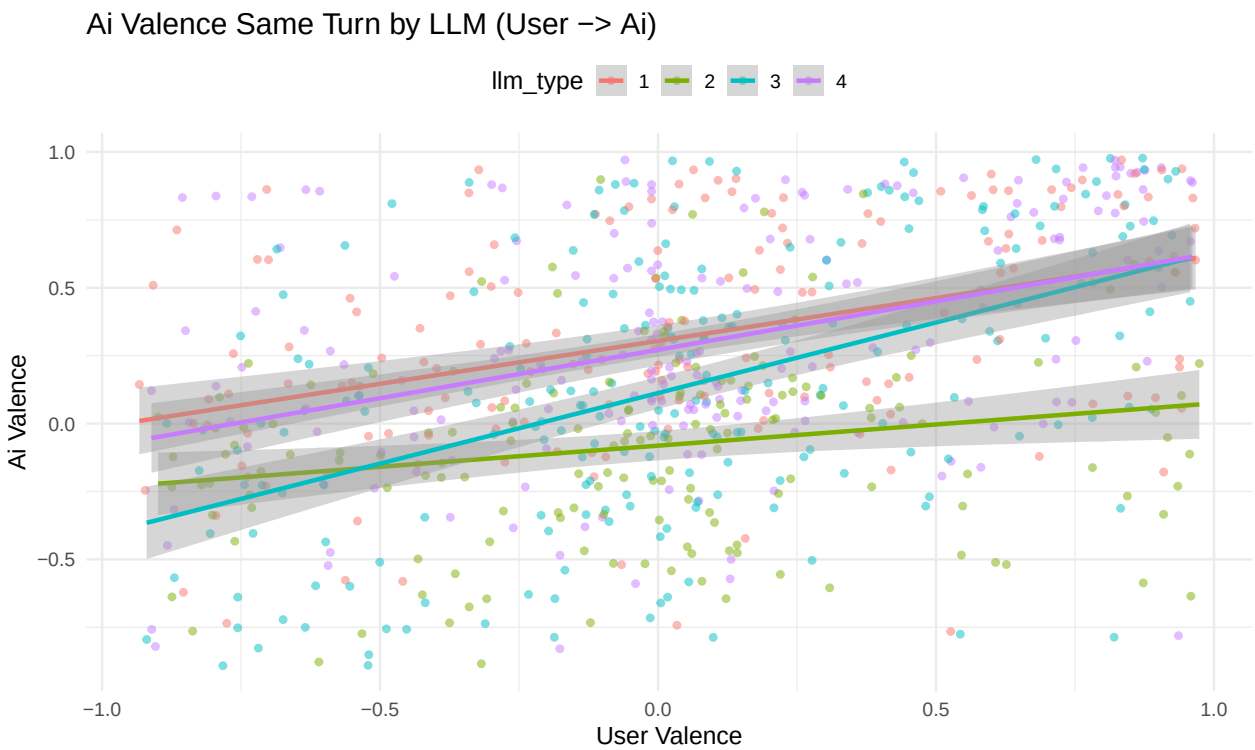
```
##
##  Welch Two Sample t-test
##
## data:  rho by starter
## t = 0.67845, df = 76.928, p-value = 0.4995
## alternative hypothesis: true difference in means between group ai and group user is not equal to 0
## 95 percent confidence interval:
##  -0.1032961  0.2100598
## sample estimates:
##   mean in group ai mean in group user
##          0.2854211          0.2320392
```

6.2 6.2 LLM Type Interaction Effects

6.2.1 6.2.1 Same-Turn Alignment by LLM Type

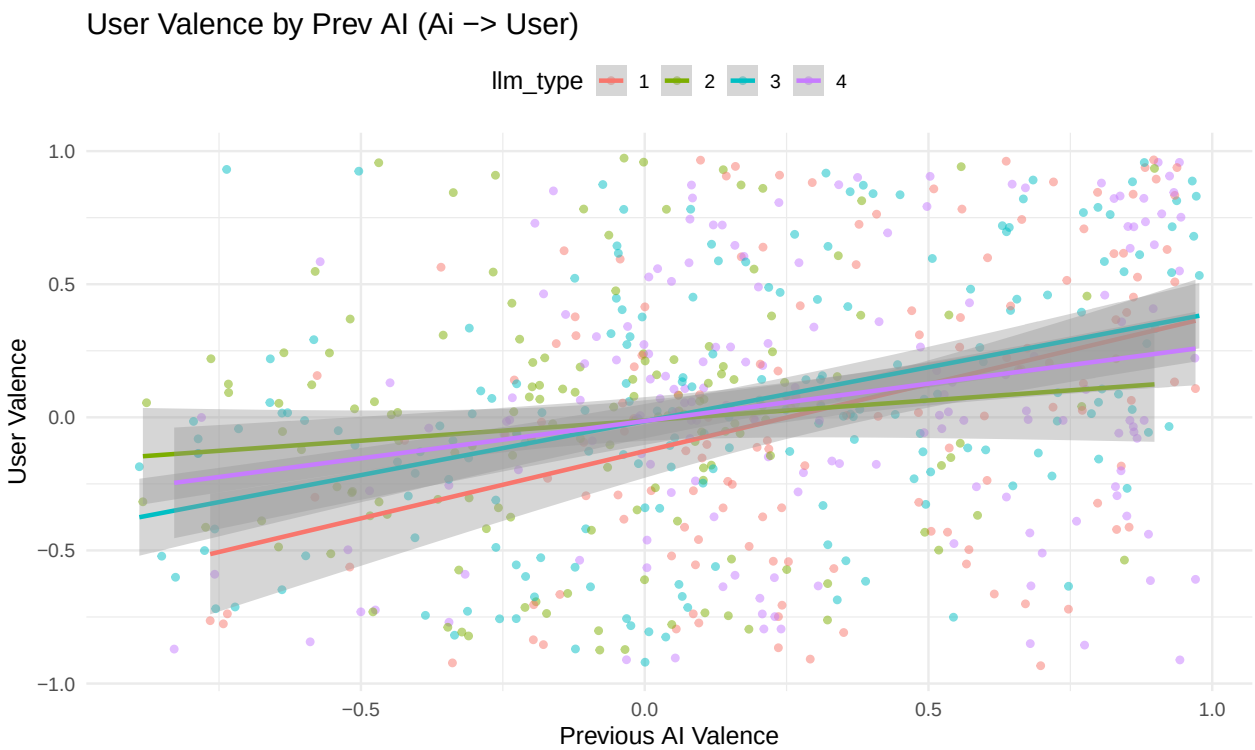
```
## === Same-Turn Alignment: LLM Type Interaction ===
```

```
##                                   Estimate Std. Error        df    t value
## (Intercept)                     0.30749849 0.04841450  70.92030  6.3513712
## user_valence                    0.25232306 0.05778417 715.62850  4.3666471
## factor(llm_type)2              -0.38954160 0.06841356  70.71099 -5.6939236
## factor(llm_type)3              -0.18848685 0.06405540  70.94386 -2.9425599
## factor(llm_type)4              -0.03201439 0.06608982  71.34474 -0.4844073
## user_valence:factor(llm_type)2 -0.13564995 0.08576450 705.21645 -1.5816561
## user_valence:factor(llm_type)3  0.13143140 0.08181536 720.99693  1.6064391
## user_valence:factor(llm_type)4  0.05562886 0.08231637 720.20276  0.6757934
##                                    Pr(>|t|)
## (Intercept)                     1.780284e-08
## user_valence                    1.447953e-05
## factor(llm_type)2               2.629017e-07
## factor(llm_type)3               4.395254e-03
## factor(llm_type)4               6.295806e-01
## user_valence:factor(llm_type)2 1.141765e-01
## user_valence:factor(llm_type)3 1.086153e-01
## user_valence:factor(llm_type)4 4.993887e-01
```



6.2.2 6.2.2 Previous-Turn Alignment by LLM Type

```
## === Previous-Turn Alignment: LLM Type Interaction ===

##                                     Estimate Std. Error        df    t value
## (Intercept)                       -0.10559137 0.05839554 109.71127 -1.8082095
## ai_valence_lag1                    0.43466652 0.09841888 600.81858  4.4164953
## factor(llm_type)2                  0.08917752 0.07740752  90.48489  1.1520525
## factor(llm_type)3                  0.10200619 0.07342726  93.80544  1.3892141
## factor(llm_type)4                  0.10618528 0.07881067 106.35765  1.3473466
## ai_valence_lag1:factor(llm_type)2 -0.33368610 0.14601130 626.94464 -2.2853443
## ai_valence_lag1:factor(llm_type)3 -0.10680849 0.12268177 558.93158 -0.8706142
## ai_valence_lag1:factor(llm_type)4 -0.20312504 0.13118473 596.77092 -1.5483894
##                                       Pr(>|t|)
## (Intercept)                       7.331332e-02
## ai_valence_lag1                   1.189796e-05
## factor(llm_type)2                 2.523343e-01
## factor(llm_type)3                 1.680567e-01
## factor(llm_type)4                 1.807320e-01
## ai_valence_lag1:factor(llm_type)2 2.262564e-02
## ai_valence_lag1:factor(llm_type)3 3.843386e-01
## ai_valence_lag1:factor(llm_type)4 1.220587e-01
```



6.2.3 6.2.3 Per-Story Correlations by LLM Type

```
## === Same-Turn Correlations by LLM Type ===
```

```
## # A tibble: 4 x 4
##   llm_type mean_rho sd_rho n_stories
##      <int>    <dbl>  <dbl>     <int>
## 1        1    0.261  0.290        18
## 2        2    0.147  0.336        18
## 3        3    0.312  0.328        24
## 4        4    0.292  0.434        21
```

```
##
## ANOVA (LLM type effect on same-turn rho):
```

```
##                  Df Sum Sq Mean Sq F value Pr(>F)
## factor(llm_type)  3  0.315  0.1049   0.848  0.472
## Residuals        76  9.400  0.1237
## 1 observation deleted due to missingness
```

6.3 6.3 Valence Distribution by Conditions

6.3.1 6.3.1 Valence by Starter

```
## # A tibble: 4 x 4
##   starter agent        mean_valence sd_valence
##   <chr>   <chr>               <dbl>      <dbl>
## 1 ai      ai_valence         0.166       0.440
## 2 ai      user_valence       0.0311      0.470
## 3 user    ai_valence         0.176       0.480
## 4 user    user_valence       0.0543      0.514
```

6.3.2 6.3.2 Valence by LLM Type

```
## # A tibble: 8 x 5
##   llm_type agent        mean_valence sd_valence n_turns
##      <int> <chr>               <dbl>      <dbl>   <int>
## 1        1 ai_valence         0.320       0.408     162
## 2        1 user_valence       0.0515      0.530     162
## 3        2 ai_valence        -0.0842      0.355     162
## 4        2 user_valence      -0.0183      0.469     162
## 5        3 ai_valence         0.138       0.501     216
## 6        3 user_valence       0.0482      0.475     216
## 7        4 ai_valence         0.300       0.433     189
## 8        4 user_valence       0.0807      0.495     189
```

6.4 7. Session Information

```
## R version 4.4.2 (2024-10-31)
## Platform: aarch64-apple-darwin20
## Running under: macOS Sequoia 15.6.1
##
## Matrix products: default
## BLAS:   /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/lib/libRblas.0.dylib
## LAPACK: /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/lib/libRlapack.dylib;  LAPACK v
##
## locale:
## [1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8
##
## time zone: Europe/Copenhagen
## tzcode source: internal
##
## attached base packages:
## [1] stats     graphics  grDevices utils     datasets  methods   base
##
## other attached packages:
##  [1] here_1.0.2      afex_1.5-0      cowplot_1.2.0   patchwork_1.3.2
##  [5] zoo_1.8-14      stringi_1.8.7   arrow_22.0.0    lmerTest_3.1-3
##  [9] lme4_1.1-37     Matrix_1.7-1    lubridate_1.9.4 forcats_1.0.1
## [13] stringr_1.6.0   dplyr_1.1.4     purrr_1.2.0     readr_2.1.6
## [17] tidyr_1.3.1     tibble_3.3.0    ggplot2_4.0.1   tidyverse_2.0.0
##
## loaded via a namespace (and not attached):
##  [1] tidyselect_1.2.1    farver_2.1.2        S7_0.2.1
##  [4] fastmap_1.2.0       TH.data_1.1-2       pacman_0.5.1
##  [7] digest_0.6.39       timechange_0.3.0    estimability_1.5.1
## [10] lifecycle_1.0.4     survival_3.7-0      magrittr_2.0.4
## [13] compiler_4.4.2      rlang_1.1.6         tools_4.4.2
## [16] utf8_1.2.6          yaml_2.3.10         knitr_1.50
## [19] labeling_0.4.3      bit_4.6.0           plyr_1.8.9
## [22] RColorBrewer_1.1-3  abind_1.4-8         multcomp_1.4-26
## [25] withr_3.0.2         numDeriv_2016.8-1.1 grid_4.4.2
## [28] xtable_1.8-4        emmeans_2.0.0       scales_1.4.0
## [31] MASS_7.3-61         tinytex_0.58        cli_3.6.5
## [34] mvtnorm_1.3-3       crayon_1.5.3        rmarkdown_2.30
## [37] ragg_1.5.0          reformulas_0.4.2    generics_0.1.4
## [40] rstudioapi_0.17.1   reshape2_1.4.5      tzdb_0.5.0
```

```
## [43] minqa_1.2.8         splines_4.4.2       assertthat_0.2.1
## [46] parallel_4.4.2      vctrs_0.6.5         boot_1.3-31
## [49] sandwich_3.1-1      carData_3.0-5       car_3.1-3
## [52] hms_1.1.4           bit64_4.6.0-1       Formula_1.2-5
## [55] systemfonts_1.3.1   glue_1.8.0          nloptr_2.2.1
## [58] codetools_0.2-20    gtable_0.3.6        pillar_1.11.1
## [61] htmltools_0.5.8.1   R6_2.6.1            textshaping_1.0.4
## [64] Rdpack_2.6.4        rprojroot_2.1.1     vroom_1.6.6
## [67] evaluate_1.0.5      lattice_0.22-6      rbibutils_2.4
## [70] Rcpp_1.1.0          coda_0.19-4.1       nlme_3.1-166
## [73] mgcv_1.9-1          xfun_0.54           pkgconfig_2.0.3
```

Analysis complete. All figures saved to `analysis/figures/` and tables to `analysis/tables/`.